

INTERNATIONAL ISLAMIC UNIVERSITY CHITTAGONG

Department of Computer Science and Engineering

CSE-2340: Software Development 1

Tailwind CSS v4

Week 4 · Class 3 (teaching) + Class 4 (practice) · Setup, spacing, type, responsive flex and grid

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Recall & Today's Outcome

? Recall from C1 and C2

- Which position value pins an element at a threshold?
- An absolute element measures from what?
- Why does a sticky header need an opaque background?
- How many breakpoints did we use in Week 3?

✓ By the end of today you can

- Set up Tailwind v4 three different ways
- Read the spacing and colour scales
- Build flex and grid layouts with utilities
- Use responsive prefixes correctly
- Style hover and focus states
- Rebuild a page you already wrote, faster

What Utility-First Means

WHAT YOU WROTE IN WEEK 3

```
.card {  
  background-color: #FFFFFF;  
  border: 1px solid #D6DEE6;  
  border-radius: 8px;  
  padding: 20px;  
}  
  
/* plus a class in the HTML */
```

THE SAME THING IN TAILWIND

```
<article class="bg-white  
  border border-iiuc-line  
  rounded-lg p-5">  
  ...  
</article>  
  
<!-- no CSS file at all -->
```



Every class is one CSS declaration you already know

- p-5 is padding: 1.25rem. flex is display: flex. justify-between is justify-content: space-between.
- You are not learning new CSS today. You are learning shorter names for the CSS you learned in Weeks 2 and 3.
- That is also why this week depends on the last two: if Flexbox is still shaky, the class names will feel random.

The Honest Trade-off

What you gain

- No naming things. No `.card-wrapper-inner-2`
- No switching between two files
- Consistent spacing, because the scale is fixed
- Unused classes are never generated
- Every project that uses it looks familiar

What you lose

- The HTML gets long and noisy
- You must know the class names
- Repeating the same 10 classes on 10 cards
- Harder to read at a glance than one good class name

Both are true. Learn it because the industry uses it, not because it is objectively better

- The repetition problem disappears in Week 11, when a card becomes a React component written once and reused.
- Handwritten CSS is not obsolete. You will meet both in real codebases.

Setting Up Tailwind v4

1 Browser script

- One script tag. Compiles in the browser.
- Learning and today's class

2 CLI

- npm install, one watch command, outputs a CSS file.
- A plain HTML project

3 Vite plugin

- npm create vite, then @tailwindcss/vite.
- Mini Project 2 and React later

```
<!-- Option 1, what we use today: no install, works with Live Server -->
<script src="https://cdn.jsdelivr.net/npm/@tailwindcss/browser@4"></script>

<style type="text/tailwindcss">
  @theme { --color-iiuc-navy: #003366; }
</style>
```

The browser script is for learning only. Mini Project 2 must use Option 3.

A Warning About Tutorials

Tailwind v3 — outdated

- A `tailwind.config.js` file
- `@tailwind base; @tailwind components; @tailwind utilities;`
- The old `cdn.tailwindcss.com` script
- A content array listing your files

Tailwind v4 — current

- No config file at all
- `@import "tailwindcss";`
- `@theme { }` for your own tokens
- Your files are detected automatically

! Almost every Tailwind video online still shows v3

Check the version before you follow anything. If you see a config file or three `@tailwind` lines, the tutorial predates v4 and the setup will not work.

The Spacing Scale

One step is 4px. This is the single most common misunderstanding: p-4 is 16px, not 4px.

p-1	4px	p-2	8px	p-3	12px	p-4	16px
p-5	20px	p-6	24px	p-8	32px	p-10	40px

```
<!-- The same scale drives padding, margin and gap -->
```

p-4	padding on all sides	px-5	left and right	py-3	top and bottom
m-4	margin on all sides	mt-8	top only	mx-auto	centre it
gap-5	gap between flex/grid items				

Because everyone uses the same steps, spacing stays consistent without anyone deciding on 23px.

Text and Colour

```
<!-- TEXT -->
text-sm text-base text-lg text-xl text-2xl
font-normal font-bold
leading-relaxed          (line-height)
text-center text-left
underline no-underline

<!-- COLOUR: name + shade 50 to 950 -->
text-slate-500          bg-white
bg-slate-100            border-slate-300
bg-black/55            (55% opacity)
```

Shades run 50 to 950

- 50 is nearly white, 950 nearly black
- 500 is the middle
- slate, gray, sky, amber, red and more

! @theme is Week 3's :root

- Same idea as your custom properties
- The difference: the token name becomes a class automatically

```
@theme {
  --color-iiuc-navy: #003366;
}

/* now available as
bg-iiuc-navy
text-iiuc-navy
border-iiuc-navy */
```

Flexbox and Grid as Utilities

`display: flex` → **`flex`**

`flex-direction: column` → **`flex-col`**

`justify-content: space-between` → **`justify-between`**

`align-items: center` → **`items-center`**

`flex-wrap: wrap` → **`flex-wrap`**

`gap: 1.25rem` → **`gap-5`**

`display: grid` → **`grid`**

`grid-template-columns: repeat(3, 1fr)` → **`grid-cols-3`**

```
<header class="flex
flex-col gap-2
md:flex-row
md:items-center
md:justify-between">
```



Nothing new here

- Same properties as Week 3
- Same axes, same rules
- If a layout does not work, the Flexbox is wrong, not the Tailwind

Responsive Prefixes

```
<!-- No prefix = ALL screens, starting from the phone -->
<div class="grid grid-cols-1 gap-5
      md:grid-cols-2
      lg:grid-cols-3">
```

(none)	all screens	<i>the phone layout</i>
---------------	-------------	-------------------------

sm:	640px and up	<i>rarely needed</i>
------------	--------------	----------------------

md:	768px and up	<i>tablet</i>
------------	--------------	---------------

lg:	1024px and up	<i>laptop</i>
------------	---------------	---------------

xl:	1280px and up	<i>large screens</i>
------------	---------------	----------------------



Every prefix is min-width

- Tailwind is mobile-first and gives you no choice
- This is exactly the habit from Week 3, now enforced by the tool

The classic mistake

- Writing md:flex-col expecting a column on a phone
- It does the opposite: row on the phone, column from 768px up

States and Positioning Utilities

```
<!-- STATES: same idea as pseudo-classes -->
<a class="text-sky-200
      hover:text-iiuc-amber
      focus:text-iiuc-amber
      focus:outline-3
      focus:outline-iiuc-amber
      focus:outline-offset-2">

<!-- POSITIONING: straight from C1 -->
<header class="sticky top-0 z-10 bg-iiuc-navy">
<article class="relative overflow-hidden">
<span class="absolute top-3 right-3 z-20">
<a class="fixed right-5 bottom-5">
```

The accessibility trap

- Tailwind makes it very easy to write `hover:` and stop
- A hover-only link is invisible to keyboard users
- Write both, every time, exactly as in Week 3

✓ Everything from C1 carries over

- `relative`, `absolute`, `sticky`, `fixed`
- `top-0`, `right-3`, `inset-x-0`
- `z-10`, `z-20`
- Same rules, same traps, shorter names

01

Live Build: Rebuild the Page

The same page from Week 3, rebuilt with utilities. Do not change the HTML structure.

Live Build: Shell, Header and Hero

```
<body class="bg-iiuc-bg text-iiuc-ink leading-relaxed">

  <!-- sticky top-0 z-10 = the three declarations from C1, as classes -->
  <header class="sticky top-0 z-10 bg-iiuc-navy text-white shadow-md">
    <div class="mx-auto w-11/12 max-w-5xl py-4
      flex flex-col gap-2
      md:flex-row md:items-center md:justify-between">
      <strong class="text-lg">IIUC · CSE Courses</strong>
      <nav aria-label="Main navigation">
        <ul class="flex flex-wrap gap-5 list-none">
          <li><a href="#main" class="text-sky-200 no-underline
            hover:text-iiuc-amber focus:text-iiuc-amber">Courses</a></li>
        </ul>
      </nav>
    </div>
  </header>

  <main id="main" class="mx-auto w-11/12 max-w-5xl py-8">
    <section class="rounded-lg bg-iiuc-navy2 px-6 py-10 text-white md:px-10 md:py-14">
      <h1 class="mb-3 text-2xl font-bold md:text-4xl">Computer Science & Engineering</h1>
      <p class="mb-5 max-w-prose text-sky-100">A practical, project-based curriculum.</p>
    </section>
```

Live Build: The Card Grid

```

<!-- One column on a phone, two from md, three from lg -->
<div class="grid grid-cols-1 gap-5 md:grid-cols-2 lg:grid-cols-3">

  <article class="relative overflow-hidden rounded-lg border border-iiuc-line bg-white">

    <div class="relative h-36">
      <div class="flex h-full items-center justify-center bg-slate-100 text-sm text-slate-500">
        course image
      </div>
      <!-- inset-x-0 = left: 0 AND right: 0, the full-width overlay from C1 -->
      <p class="absolute inset-x-0 bottom-0 bg-black/55 px-3 py-2 text-sm text-white">
        Software Development 1
      </p>
    </div>

    <!-- absolute measures from the article, because it is relative -->
    <span class="absolute top-3 right-3 z-20 rounded-full bg-iiuc-amber
      px-3 py-1 text-xs font-bold text-amber-950">NEW</span>

    <div class="p-5">
      <h3 class="mb-1 text-lg font-bold text-iiuc-navy">CSE-2340</h3>
      <p class="mb-2 text-sm text-slate-500">2 credits · 3rd semester</p>
    </div>
  </article>

</div>

```

Common Mistakes

p-4 is not 4px

The scale is steps of 4px, so p-4 is 16px. p-16 is 64px.

My md: class does the opposite

Every prefix is min-width. No prefix means phone; md: means 768px and up.

justify-between does nothing

The parent is not a flex or grid container. Add flex or grid first.

grid-cols-3 is ignored

Same cause: the element needs grid before grid-template-columns applies.

My custom colour class does nothing

The token is not in @theme, or it is spelled differently there.

The tutorial does not match

It is a v3 tutorial. Check for a config file or @tailwind lines.

A class that generated no CSS simply does not exist, exactly like a misspelt property. DevTools shows you nothing, which is the clue.

02

Class 4 — Practice

Reproduce a page, make it responsive, modify it, then fix six broken class attributes.

Class 4: What You Will Do

1 Reproduce

- Build the page with utilities only
- 12 min

2 Make it responsive

- Three widths, correct prefixes
- 8 min

3 Modify it

- A theme token, both states, a new section
- 8 min

4 Fix the broken page

- Six bugs, all in class attributes
- 8 min

! Mini Project 2 is due Week 5, C4

Your own responsive design, built with Tailwind, deployed. Everything today is directly reusable in it, and MP2 must use the Vite setup, not the browser script.

Exercise 1: Reproduce the Page

Build with utilities only

- Sticky header, opaque, with nav links
- Centred container: mx-auto, max-width, page padding
- Three course cards in a grid
- A badge positioned on each card corner



Use the scale

- Padding and gaps from the 4px scale: p-4, p-5, gap-5
- Colours from the 50-950 range
- No arbitrary values unless nothing on the scale fits

```
<!-- Start from this shell -->
<div class="mx-auto w-11/12 max-w-5xl py-8">
  <div class="grid gap-5">    <!-- add the column classes in Exercise 2 -->
```

Exercise 2: Make It Responsive

1 360px

One column. Header stacked. No horizontal scrollbar

2 768px

Two columns. Header becomes a row, spread apart

3 1024px

Three columns. Larger page heading



The unprefixed class is the phone

`grid-cols-1 md:grid-cols-2 lg:grid-cols-3`

Write it in a comment: which class does which width?

Exercise 3: Modify It

1 Add a theme token

Define `--color-iiuc` in `@theme` and use `bg-iiuc` on the header

2 Add both states

`hover:` and `focus:` on every link and button. Test with Tab only

3 Add a new section

A two-column block: text on the left, a placeholder on the right at md and up

```
<style type="text/tailwindcss">@theme { --color-iiuc: #003366; }</style>
```

Exercise 4: Fix the Broken Page

Open starter/. Six bugs, all in the class attributes. No CSS file is involved.

1 The cards never become a row

Read the prefix. Which width is it for?

3 grid-cols-3 is ignored

Same family of problem as bug 2

5 Keyboard users get no feedback

One prefix is missing

2 gap and justify do nothing

Is the container a flex container at all?

4 The heading colour does not apply

Does that colour token exist?

6 Horizontal scrollbar at 360px

A fixed width where a responsive one belongs

In DevTools, select the element and read its class list. A class that generated no CSS simply does not exist.

Assignment - Week 4, C3 and C4

1 Part 1 - In-class demonstration

- Time: last 5 minutes of C4
- Task: show your Tailwind page at 360px, 768px and 1024px
- Name the class that controls each breakpoint
- Tab to a link and show the focus style
- Marks: 2 (checked on the spot)

2 Part 2 - Home task

- Finish all three modifications and all six bug fixes
- Rebuild ONE page from Week 3 using Tailwind, set up with Vite this time
- Write tailwind-notes.md: three utilities you had to look up, and what they do
- Commit and push
- Deadline: before Week 5, Class 1 · Marks: 3

Mini Project 2 is due Week 5, C4: your own responsive design in Tailwind, deployed. Start it this weekend.

Marking Rubric & Exit Check

RUBRIC - 5 MARKS TOTAL

Page reproduced with utilities only	1.5
Correct responsive behaviour at all three widths	1.5
All six planted bugs fixed	1.0
Week 3 page rebuilt with a Vite setup	1.0

*Next week: responsive deployment, image optimisation and JavaScript fundamentals.
MP2 is evaluated in C4.*

Exit question 1 - concept

What does `md:flex-row` say about how the page looks on a phone?

Exit question 2 - practical

Write the classes for a container that is centred, 92% wide, with a maximum width.